

8. Let vectors $\vec{u} = -6\vec{i} - 78\vec{j}$ and $\vec{v} = 117\vec{i} + k\vec{j}$ such that $\vec{u} \perp \vec{v}$. Determine the value of k . Express your answer as an integer or as a common or improper fraction.

9. **(Always, Sometimes, or Never)** For your answer, write the whole word **Always**, **Sometimes**, or **Never**—whichever is correct.

The graph of a certain linear function has a negative slope. Consequently, if its x -intercept is a negative number, then its y -intercept is also a negative number.

10. Let $\sin x = \frac{1}{3} + \cos x$. Determine the value of $\sin(2x)$. Express your answer as an integer or as a common or improper fraction.

11. Box A contains 3 red marbles and 6 blue marbles. Box B contains 2 red marbles and 1 blue marble. Box C contains 2 red marbles and 4 blue marbles. Two of the boxes are randomly chosen. Then one marble is chosen from each of those two boxes. Determine the probability that these two marbles drawn are the same color. Express your answer as a common fraction.

12. Consider the graph of the function $f(x) = 3\sin(2x + \pi) - 5$. Determine the amplitude of $f(x)$.

13. Let x represent the degree measure of an angle such that $\tan(x) = \sqrt{3}$. If $0^\circ < x^\circ < 720^\circ$, determine the sum of all possible distinct values of x .

14. a , b , and c are positive integers with $b < c$ such that $\frac{1}{\sqrt[3]{2}-1} = a + \sqrt[3]{b} + \sqrt[3]{c}$. Determine the ordered triple (a, b, c) . Express your answer as that ordered triple.